

COMP0086 Past Exam Questions

Common Topics with COMP0171 — Problems and Full Solutions

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1 Topic 1: Free Energy / ELBO / EM Algorithm

1.1 2017 Q1: Free Energies [20 marks]

Also appears verbatim as 2022 Q5.

In a fit of confusion, one of your colleagues is attempting to derive the expectation maximisation (EM) algorithm for a mixture model using the free energy, but has forgotten the entropy term.

That is, given data $\{x_i\}$ and a generic mixture model with M components and parameters $\pi, \theta_1, \dots, \theta_M$:

$$s_i \sim \text{Discrete}(\pi), \quad s_i \in \{1 \dots M\}; \quad \pi_m \geq 0; \quad \sum_{m=1}^M \pi_m = 1$$

$$x_i | s_i = m \sim P_m(x_i; \theta_m), \quad x_i \in \mathbb{R}^D,$$

he has introduced posterior responsibilities $r_i^m \geq 0$ with $\sum_{m=1}^M r_i^m = 1$ for each observation i , and is seeking to maximise

$$\tilde{\mathcal{F}}(\{r_i^m\}, \{\theta_m\}, \pi) = \sum_i \sum_m r_i^m \log P(x_i, s_i = m | \theta_m, \pi),$$

with respect to $\{r_i^m\}$, $\{\theta_m\}$ and π .

However, he quickly runs into a problem. When using the method of Lagrange multipliers to find the optima with respect to r_i^m (as we did in lecture) he obtains derivatives in which the r_i^m do not appear. He calls you over to help.

(a) You suggest thinking geometrically. Taking the case $M = 2$, sketch the isovalue contours for $\tilde{\mathcal{F}}$ as a function of the responsibilities for the first data point, as well as the constraint surface $\sum_m r_1^m = 1$. Argue that the maximum will typically be found when $r_1^m = 1$ for some m . What determines which m ? [5 marks]

Solution

For $M = 2$, the responsibilities satisfy $r_1^1 + r_1^2 = 1$, so the constraint surface is a line segment from $(1, 0)$ to $(0, 1)$.

The objective $\tilde{\mathcal{F}}$ depends on (r_1^1, r_1^2) through the term

$$r_1^1 \log P(x_1, s_1=1 | \theta_1, \pi) + r_1^2 \log P(x_1, s_1=2 | \theta_2, \pi).$$

This is *linear* in (r_1^1, r_1^2) since the log-joint terms do not depend on r_1^m . The isovalue contours are therefore straight parallel lines.

A linear function on a convex polytope is maximised at a vertex. The simplex $\{r_1^1 + r_1^2 = 1, r_1^m \geq 0\}$ has vertices $(1, 0)$ and $(0, 1)$. Therefore the maximum is attained at $r_1^m = 1$ for exactly one m :

$$r_1^{m^*} = 1 \quad \text{where} \quad m^* = \arg \max_m \log P(x_1, s_1=m | \theta_m, \pi).$$

The component with the larger joint probability receives all the responsibility (hard assignment).

(b) Derive the correct free energy for this model, showing explicitly that it lower bounds the log-likelihood. You should start from the log-likelihood for the model above, and complete the derivation using the specific model notation — do **not** reproduce a generic derivation and substitute the model-specific values at the final step. [5 marks]

Solution

The log-likelihood is:

$$\ell(\theta, \pi) = \sum_i \log \sum_{m=1}^M P(x_i, s_i=m | \theta_m, \pi).$$

Introduce responsibilities $r_i^m \geq 0$, $\sum_m r_i^m = 1$, and multiply and divide inside the log:

$$\ell = \sum_i \log \sum_{m=1}^M r_i^m \frac{P(x_i, s_i=m | \theta_m, \pi)}{r_i^m}.$$

Applying Jensen's inequality (log is concave, r_i^m are non-negative weights summing to 1):

$$\begin{aligned} \ell &\geq \sum_i \sum_m r_i^m \log \frac{P(x_i, s_i=m | \theta_m, \pi)}{r_i^m} \\ &= \underbrace{\sum_i \sum_m r_i^m \log P(x_i, s_i=m | \theta_m, \pi)}_{\tilde{\mathcal{F}}} - \underbrace{\sum_i \sum_m r_i^m \log r_i^m}_{H_q \geq 0} \equiv \mathcal{F}. \end{aligned}$$

The correct free energy is $\mathcal{F} = \tilde{\mathcal{F}} + H_q$, and $\mathcal{F} \leq \ell$ by Jensen.

(c) Show that the incorrect free energy $\tilde{\mathcal{F}}$ is also a lower bound on the log likelihood. [3 marks]

Solution

Since $r_i^m \in [0, 1]$, we have $\log r_i^m \leq 0$, so $H_q = -\sum_i \sum_m r_i^m \log r_i^m \geq 0$.

From (b): $\tilde{\mathcal{F}} = \mathcal{F} - H_q \leq \mathcal{F} \leq \ell$.

Therefore $\tilde{\mathcal{F}}$ is also a lower bound on ℓ , though a looser one than $\mathcal{F}$.

(d) Now consider a continuum of free energies $\mathcal{F}_\tau$, $\tau \in [0, 1]$ in which the entropy term is premultiplied by τ . Derive the optimal responsibilities for $\mathcal{F}_\tau$ when $\tau > 0$. Then take the limit $\tau \rightarrow 0$ and show that you obtain the same result as you found geometrically. [5 marks]

Solution

Define $\mathcal{F}_\tau = \sum_i \sum_m r_i^m \log P(x_i, s_i=m) - \tau \sum_i \sum_m r_i^m \log r_i^m$.

For $\tau > 0$, optimise w.r.t. r_i^m subject to $\sum_m r_i^m = 1$ via Lagrange multipliers:

$$\frac{\partial \mathcal{L}}{\partial r_i^m} = \log P(x_i, s_i=m) - \tau(\log r_i^m + 1) - \lambda = 0 \implies r_i^m \propto P(x_i, s_i=m)^{1/\tau}.$$

Normalising:

$$r_i^m = \frac{P(x_i, s_i=m)^{1/\tau}}{\sum_{m'} P(x_i, s_i=m')^{1/\tau}}.$$

As $\tau \rightarrow 0$, the exponent $1/\tau \rightarrow \infty$ and $P^{1/\tau}$ is dominated by the largest term:

$$\lim_{\tau \rightarrow 0} r_i^m = \begin{cases} 1 & \text{if } m = \arg \max_{m'} P(x_i, s_i=m'), \\ 0 & \text{otherwise.} \end{cases}$$

This is the same hard assignment found geometrically in (a). The parameter τ acts as an inverse temperature: $\tau = 1$ gives standard EM soft assignments, $\tau \rightarrow 0$ gives hard assignments.

(e) What well-known algorithm will result if you maximise $\tilde{\mathcal{F}}$ for a mixture of Gaussians with fixed identity covariance matrices and unknown means, and fixed mixing proportions $\pi_m = 1/M$? [2 marks]

Solution

With identity covariances, uniform mixing, and hard assignments, the E-step assigns each x_i to $m^* = \arg \min_m \|x_i - \mu_m\|^2$, and the M-step sets μ_m to the mean of assigned points.

This is exactly the **K-means** algorithm.

1.2 2019 Q3: Free-Energy Derivatives [20 marks]

Also appears verbatim as 2022 Q3.

Consider a general latent variable model with observed data $\mathcal{X}$, latent variables $\mathcal{Z}$ and parameters θ . The log-likelihood function is given by

$$\ell(\theta) = \log \int d\mathcal{Z} P(\mathcal{X}, \mathcal{Z} | \theta).$$

(a) How is the free energy defined? Show that it is a lower bound on the log likelihood. [4 marks]

Solution

The free energy is:

$$\mathcal{F}(q, \theta) = \int d\mathcal{Z} q(\mathcal{Z}) \log \frac{P(\mathcal{X}, \mathcal{Z} | \theta)}{q(\mathcal{Z})} = \langle \log P(\mathcal{X}, \mathcal{Z} | \theta) \rangle_q + H_q.$$

Rewriting:

$$\mathcal{F}(q, \theta) = \ell(\theta) - D_{\text{KL}}[q(\mathcal{Z}) \| P(\mathcal{Z} | \mathcal{X}, \theta)].$$

Since $D_{\text{KL}} \geq 0$, we have $\mathcal{F}(q, \theta) \leq \ell(\theta)$, with equality iff $q = P(\mathcal{Z} | \mathcal{X}, \theta)$.

(b) Show that the (generalised) M-step gradient evaluated after an exact E-step is (under some regularity conditions) equal to the gradient of log likelihood with respect to the parameters. [6 marks]

Solution

From (a): $\ell(\theta) = \mathcal{F}(q, \theta) + D_{\text{KL}}[q \| P(\mathcal{Z} | \mathcal{X}, \theta)]$, so

$$\frac{\partial \ell}{\partial \theta} = \frac{\partial \mathcal{F}}{\partial \theta} + \frac{\partial}{\partial \theta} D_{\text{KL}}[q \| P(\mathcal{Z} | \mathcal{X}, \theta)].$$

After an exact E-step, $q = P(\mathcal{Z} | \mathcal{X}, \theta^{(k-1)})$, so $D_{\text{KL}} = 0$ at $\theta = \theta^{(k-1)}$. Since $D_{\text{KL}} \geq 0$ and equals zero there, this is a minimum w.r.t. θ , so (under regularity conditions):

$$\frac{\partial}{\partial \theta} D_{\text{KL}} \Big|_{\theta^{(k-1)}} = 0.$$

Since H_q does not depend on θ : $\frac{\partial \mathcal{F}}{\partial \theta} = \frac{\partial}{\partial \theta} \langle \log P(\mathcal{X}, \mathcal{Z} | \theta) \rangle_q$. Therefore:

$$\frac{\partial}{\partial \theta} \Big|_{\theta^{(k-1)}} \langle \log P(\mathcal{X}, \mathcal{Z} | \theta) \rangle_{P(\mathcal{Z} | \mathcal{X}, \theta^{(k-1)})} = \frac{\partial \ell}{\partial \theta} \Big|_{\theta^{(k-1)}}.$$

(c) Consider a mixture of spherical Gaussians:

$$p(z_n) = \text{Discrete}(\pi), \quad p(x_n | z_n) = \prod_k (\mathcal{N}(\mu_k, I))^{\delta_{z_n, k}}$$

where $z_n \in 1 \dots K$ and $\delta_{z_n, k}$ is an indicator.

Compute the exact M-step update for μ_k , the gradient, and show the exact step changes μ_k along the gradient direction. [6 marks]

Solution

The expected complete-data log-likelihood (terms in μ_k) is:

$$\langle \log P \rangle_q = \sum_n r_{nk} \left[-\frac{1}{2} \|x_n - \mu_k\|^2 \right] + \text{const}, \quad r_{nk} = P(z_n = k | x_n, \theta^{\text{old}}).$$

Gradient: $\frac{\partial}{\partial \mu_k} \langle \log P \rangle_q = \sum_n r_{nk} (x_n - \mu_k)$.

Exact M-step: Setting to zero: $\mu_k^{\text{new}} = \frac{\sum_n r_{nk} x_n}{\sum_n r_{nk}}$.

Alignment:

$$\Delta\mu_k = \mu_k^{\text{new}} - \mu_k^{\text{old}} = \frac{\sum_n r_{nk}(x_n - \mu_k^{\text{old}})}{\sum_n r_{nk}} = \frac{1}{\sum_n r_{nk}} \frac{\partial}{\partial \mu_k} \langle \log P \rangle_q.$$

Since $\sum_n r_{nk} > 0$, $\Delta\mu_k$ is a positive scalar multiple of the gradient.

(d) Is this generally true? That is, is the change in parameters resulting from an exact (i.e. non-generalised) M-step always aligned with the gradient direction? Explain. [4 marks]

Solution

No. Here the expected log-likelihood is concave quadratic in μ_k , forcing the closed-form maximiser to lie along the gradient from the current point. For general parameters (e.g. covariance matrices, constrained parameters), the M-step maximises globally while the gradient is only the local linear approximation. These can disagree when the objective is not quadratic or the parameter space has curvature.

2 Topic 2: Conjugate Priors / Bayesian Linear Regression

2.1 2018 Q1: Conjugate Mines [20 marks]

Also appears verbatim as 2020 Q1.

(a) Define a conjugate prior.

[2 marks]

Solution

A family of prior distributions Π is *conjugate* to a family of likelihood functions $\{p(x | \theta)\}$ if, for every prior $\pi \in \Pi$, the posterior $\pi(\theta | x) \propto p(x | \theta)\pi(\theta)$ is also in Π . The posterior has the same functional form as the prior, differing only in the hyperparameters.

(b) How are the parameters of the posterior distribution related to those of the conjugate prior? [2 marks]

Solution

For exponential family likelihoods, the posterior hyperparameters are: $\tau \rightarrow \tau + \sum_i T(x_i)$ and $\nu \rightarrow \nu + n$, i.e. the data sufficient statistics are added to the prior pseudo-sufficient-statistics, and the pseudo-count is incremented by the sample size.

Suppose that a rather devious type of mine contains an internal clock that ticks with period θ . The delay t between arming and explosion is $t \sim \text{Uniform}(0, \theta)$. You arm n mines and observe delays $t_1, \dots, t_n$. Assume all mines share the same θ .

(c) What is the likelihood function for θ ? What is the MLE? Is it reasonable for small n ? [2 marks]

Solution

$L(\theta) = \prod_i \frac{1}{\theta} \mathbf{1}[t_i \leq \theta] = \theta^{-n} \mathbf{1}[\theta \geq t_{\max}]$, where $t_{\max} = \max_i t_i$.

This decreases for $\theta \geq t_{\max}$, so $\hat{\theta}_{\text{ML}} = t_{\max}$.

Not reasonable for small n : it is biased downward ($t_{\max} \leq \theta$ always). Indeed $\mathbb{E}[\hat{\theta}] = \frac{n}{n+1}\theta$.

(d) Show that a conjugate prior has the Pareto form $p(\theta) = \frac{1}{Z(\theta_0, k)} \Theta(\theta - \theta_0) \theta^{-k}$. Calculate $Z(\theta_0, k)$. [3 marks]

Solution

The likelihood is $L(\theta) \propto \theta^{-n} \Theta(\theta - t_{\max})$. With prior $p(\theta) \propto \Theta(\theta - \theta_0) \theta^{-k}$, the posterior is:

$$p(\theta | t_{1:n}) \propto \theta^{-(n+k)} \Theta(\theta - \max(\theta_0, t_{\max})),$$

which is Pareto with $k' = k + n$, $\theta'_0 = \max(\theta_0, t_{\max})$: same form, confirming conjugacy.

Normalising constant (for $k > 1$):

$$Z(\theta_0, k) = \int_{\theta_0}^{\infty} \theta^{-k} d\theta = \frac{\theta_0^{-(k-1)}}{k-1}.$$

(e) Give the normalised posterior on θ and the MAP estimate.

[2 marks]

Solution

$p(\theta | t_{1:n}) = \text{Pareto}(\theta'_0, k')$ with $k' = k + n$, $\theta'_0 = \max(\theta_0, t_{\max})$:

$$p(\theta | t_{1:n}) = (k' - 1)(\theta'_0)^{k'-1} \theta^{-k'} \Theta(\theta - \theta'_0).$$

The Pareto density is monotonically decreasing, so $\hat{\theta}_{\text{MAP}} = \theta'_0 = \max(\theta_0, t_{\max})$.

Now you wonder whether all clocks share the same period. Assume reasonable θ_0 and k .

(f) What is the marginal likelihood under the single- θ model? [3 marks]

Solution

$$P(t_{1:n}) = \int_{\theta'_0}^{\infty} \theta^{-n} \cdot \frac{(k-1)\theta_0^{k-1}}{\theta^k} d\theta = \frac{k-1}{n+k-1} \cdot \frac{\theta_0^{k-1}}{(\theta'_0)^{n+k-1}}, \quad \theta'_0 = \max(\theta_0, t_{\max}).$$

(g) If each clock has $\theta_i \sim_{\text{iid}} \text{Pareto}(\theta_0, k)$, what is the marginal likelihood? [3 marks]

Solution

$$P(t_{1:n}) = \prod_{i=1}^n \frac{k-1}{k} \cdot \frac{\theta_0^{k-1}}{(\max(\theta_0, t_i))^k}.$$

(h) If all delays are identical, which model is favoured (in the limit $n \gg 1$)? [3 marks]

Solution

With $t_i = t^*$ for all i , the dominant exponent for large n in the single- θ model is $\sim n$, while in the independent model it is $\sim nk$ with $k > 1$. So $P_{\text{single}} \gg P_{\text{indep}}$.
The **single- θ model is favoured**: identical delays are expected when all clocks share one period. The independent model wastes prior mass on configurations with differing θ_i .

2.2 2019 Q1: Conjugacy in Regression [20 marks]

Also appears verbatim as 2022 Q1.

Consider linear regression $y_n \sim \mathcal{N}(w^\top x_n, \sigma^2)$ for pairs (x_n, y_n) , $n = 1 \dots N$.

(a) What is a conjugate prior? [2 marks]

Solution

A conjugate prior is one whose posterior, after updating via Bayes' rule with the given likelihood, belongs to the same parametric family as the prior, differing only in the hyperparameter values.

(b) State the conjugate prior for w when σ^2 is known. [2 marks]

Solution

A Gaussian: $p(w) = \mathcal{N}(w; \mu_0, \Sigma_0)$, commonly written $\mathcal{N}(0, A^{-1})$.

Now let σ^2 be unknown with inverse-gamma prior $p(\sigma^2) = \text{InvGamma}(\sigma^2; \alpha, \beta)$. Let $X = [x_1 \dots x_N]$, $Y = [y_1 \dots y_N]$.

(c) Find the joint posterior on (σ^2, w) under independent Gaussian and inverse-gamma priors. Show it is not conjugate. [6 marks]

Solution

The joint posterior is:

$$p(w, \sigma^2 | Y) \propto (\sigma^2)^{-N/2} \exp\left(-\frac{\|Y - w^\top X\|^2}{2\sigma^2}\right) \cdot \exp\left(-\frac{1}{2}w^\top A w\right) \cdot (\sigma^2)^{-\alpha-1} e^{-\beta/\sigma^2}.$$

The w -dependent terms are $-\frac{1}{2}w^\top (A + \frac{XX^\top}{\sigma^2})w + w^\top \frac{XY^\top}{\sigma^2}$. Here A appears *without* σ^{-2} , while data terms carry σ^{-2} . The conditional on w given σ^2 has covariance not proportional to σ^2 (because of the A term). The posterior does not have the form $\text{InvGamma} \times \mathcal{N}$; the independent prior is **not conjugate**.

The correct conjugate prior is $p(\sigma^2, w) = \text{InvGamma}(\sigma^2; \alpha, \beta) \mathcal{N}(w; \mu, \sigma^2 A^{-1})$. Take $\mu = 0$.

(d) Show this is conjugate and derive the hyperparameter updates.

[8 marks]

Solution

Let $d = \dim(w)$. The prior is:

$$p(\sigma^2, w) \propto (\sigma^2)^{-\alpha-1-d/2} \exp\left(-\frac{\beta + \frac{1}{2}w^\top A w}{\sigma^2}\right).$$

All w -and- σ^2 interactions now appear as $w^\top(\cdot)w/\sigma^2$. Multiplying by the likelihood and collecting:

$$p(w, \sigma^2 | Y) \propto (\sigma^2)^{-(\alpha+N/2)-1-d/2} \exp\left(-\frac{w^\top (A + X X^\top) w - 2w^\top X Y^\top + Y Y^\top + 2\beta}{2\sigma^2}\right).$$

Setting $A' = A + X X^\top$, $\mu' = (A')^{-1} X Y^\top$, and completing the square:

$$p(w, \sigma^2 | Y) = \text{InvGamma}(\sigma^2; \alpha', \beta') \cdot \mathcal{N}(w; \mu', \sigma^2 (A')^{-1}),$$

with:

$$A' = A + X X^\top, \quad \mu' = (A')^{-1} X Y^\top, \quad \alpha' = \alpha + \frac{N}{2}, \quad \beta' = \beta + \frac{1}{2}(Y Y^\top - \mu'^\top A' \mu').$$

Same Normal-Inverse-Gamma form $\implies$ conjugacy confirmed.

(e) Would integrating out both w and σ^2 change the optimal A compared to integrating out w alone? [2 marks]

Solution

The marginal likelihood $P(Y | A)$ integrating out both is closed-form. For large N the posterior on σ^2 concentrates around its ML estimate, so the two approaches give approximately the same optimal A . For finite N with an informative prior on σ^2 , they generally differ.

3 Topic 3: Gaussian Processes

3.1 2018 Q6: Periodic Gaussian Processes [20 marks]

Recall that a GP prior $f(t) \sim \mathcal{GP}(0, K)$ can be derived from $\beta \sim \mathcal{N}(0, I)$, $f(t) = \beta^\top \phi(t)$.

(a) Derive the relationship between $\phi(t)$ and the kernel $K(t, t')$.

[4 marks]

Solution

$$K(t, t') = \text{Cov}[f(t), f(t')] = \mathbb{E}[\beta^\top \phi(t) \phi(t')^\top \beta] = \phi(t)^\top \underbrace{\mathbb{E}[\beta \beta^\top]}_I \phi(t') = \phi(t)^\top \phi(t').$$

(b) With noisy observations $y \sim \mathcal{N}(\beta^\top \phi(t), \sigma^2)$, what is the marginal covariance matrix for $\{y_i\}$ at $\{t_i\}$? [2 marks]

Solution

$$\Sigma_{ij} = \phi(t_i)^\top \phi(t_j) + \sigma^2 \delta_{ij} = K(t_i, t_j) + \sigma^2 \delta_{ij}.$$

The standard periodic kernel is $K(t, t') = s^2 \exp(-2 \sin^2(\pi|t - t'|/p)/l^2)$.

(c) Interpret the hyperparameters s , p and l .

[3 marks]

Solution

s^2 : signal variance (amplitude of draws). p : period of the function. l : length-scale within each period (smoothness of the periodic shape; small l allows rapid variation, large l gives smoother/more sinusoidal shapes).

(d) Sketch a draw from the product kernel $K = s^2 \exp(-2 \sin^2(\pi|t - t'|/p)/l^2) \exp(-(t - t')^2/(2r^2))$. [4 marks]

Solution

This is a periodic kernel multiplied by a squared-exponential with length-scale r . A draw looks like a *locally periodic* function: oscillations of period p , amplitude $\sim s$, smoothness l , but the periodic pattern gradually decorrelates on scale r . Visually: near any time the function looks periodic, but the shape drifts as you move more than r away.

An alternative periodic kernel is $K(t, t') = s^2 \cos(2\pi(t - t')/p)$.

(e) Show this is a valid kernel by constructing $\phi(t)$.

[2 marks]

Solution

Using $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$, set

$$\phi(t) = s \begin{pmatrix} \cos(2\pi t/p) \\ \sin(2\pi t/p) \end{pmatrix}.$$

Then $\phi(t)^\top \phi(t') = s^2 \cos(2\pi(t - t')/p)$.

(f) Describe the set of functions with non-zero density under this GP prior.

[3 marks]

Solution

Since ϕ is 2-dimensional, $f(t) = \beta_1 \cos(2\pi t/p) + \beta_2 \sin(2\pi t/p)$. The set of non-zero-density functions is exactly the sinusoids with period p : a two-dimensional function space.

(g) Why is the exponential form more commonly used?

[2 marks]

Solution

The cosine kernel restricts to a 2D space of pure sinusoids. The exponential periodic kernel spans an infinite-dimensional space of periodic functions with arbitrary within-period shapes. Real phenomena are rarely pure sinusoids.

3.2 2019 Q6: GPFA [20 marks]

Consider $z_k(\cdot) \sim \mathcal{GP}(\mu_k, \kappa_k)$ for $k = 1 \dots K$, and $x_d(t) \sim \mathcal{N}(\sum_k \Lambda_{dk} z_k(t), \psi_d)$. Observations form $D \times N$ matrix X ; latents form $K \times N$ matrix Z .

(a) Find the posterior on Z .

[3 marks]

Solution

The model is linear-Gaussian. Let $\Psi = \text{diag}[\psi_d]$ and K_k be the $N \times N$ GP covariance matrix. The posterior $P(Z | X)$ is Gaussian, obtained from standard Gaussian conditioning on the joint (Z, X) .

(b) Give the EM M-step for Λ (assuming μ_k, κ_k known).

[3 marks]

Solution

$$\hat{\Lambda} = \left(\sum_n x(t_n) \langle z(t_n) \rangle_q^\top \right) \left(\sum_n \langle z(t_n) z(t_n)^\top \rangle_q \right)^{-1},$$

where expectations are under the posterior from (a).

(c) Compute $P(x(t^*) | X)$.

[4 marks]

Solution

Marginalise over $z(t^*)$ using its GP predictive posterior. The result is Gaussian:

$$\mathbb{E}[x(t^*) | X] = \Lambda \mathbb{E}[z(t^*) | X], \quad \text{Cov}[x(t^*) | X] = \Lambda \text{Cov}[z(t^*) | X] \Lambda^\top + \Psi.$$

(d) Describe an approach when only $x_{d_n}(t_n)$ is observed at each time t_n .

[6 marks]

Solution

This is EM with missing data. The E-step computes the posterior on Z using only the observed $x_{d_n}(t_n)$ at each t_n (effectively selecting row d_n of Λ). The M-step updates each row d of Λ using only time points where $d_n = d$. Prediction follows (c) using the updated posterior.

(e) Under what constraints are GPFA and the LGSSM equivalent?

[4 marks]

Solution

The two are equivalent when: (i) each GP has a Markov covariance function (e.g. Matérn-1/2, admitting a finite state-space representation), (ii) observations are at regularly spaced times, and (iii) the LGSSM is stationary. The GP must have a finite-dimensional state-space form.

3.3 2020 Q6: Tracking Drones [20 marks]

A drone trajectory $x(t) \sim \text{GP}(0, K)$. Camera traps at locations $x_1, \dots, x_L$ report times of passage, but may be incomplete.

(a) For $n_l = 1$ for all l , give the predicted distribution at query time t .

[2 marks]

Solution

Standard GP prediction:

$$x(t) \mid \text{data} \sim \mathcal{N}(k(t)^\top K_{\text{obs}}^{-1} x_{\text{obs}}, K(t, t) - k(t)^\top K_{\text{obs}}^{-1} k(t)),$$

with $x_{\text{obs}} = [x_1, \dots, x_L]^\top$, $[K_{\text{obs}}]_{ll'} = K(t_{l,1}, t_{l',1})$, $k(t) = [K(t, t_{l,1})]_l$.

(b) How does the answer change for arbitrary n_l ?

[2 marks]

Solution

The observation set expands to $N = \sum_l n_l$ pairs. The same GP prediction formula applies with the enlarged observation vector and kernel matrix.

(c) Would the expression be valid if traps were completely reliable?

[4 marks]

Solution

No. Reliable traps report *all* passage times, so absence of a time means $x(t) \neq x_l$. This negative information introduces inequality constraints, making the posterior non-Gaussian. With unreliable traps, absence is uninformative, so conditioning on positives only is valid.

Now consider the velocity $v(t) = dx/dt$.

(d) Find $K_{xv}(t, t')$ between $x(t)$ and $v(t')$.

[4 marks]

Solution

$$K_{xv}(t, t') = \text{Cov}[x(t), v(t')] = \text{Cov}[x(t), \frac{dx}{dt'}(t')] = \frac{\partial K(t, t')}{\partial t'}.$$

(e) Give the predicted distribution of $v(t)$ given trap times.

[4 marks]

Solution

Multi-output GP prediction with cross-covariances K_{xv}, K_{vx}, K_{vv} :

$$v(t) \mid \text{data} \sim \mathcal{N}(k_{vx}(t)^\top K_{\text{obs}}^{-1} x_{\text{obs}}, K_{vv}(t, t) - k_{vx}(t)^\top K_{\text{obs}}^{-1} k_{xv}(t)),$$

where $K_{vv}(t, t') = \frac{\partial^2 K}{\partial t \partial t'}$, $k_{vx}(t) = [\partial_t K(t, t_l)]_l$, $k_{xv}(t) = [\partial_{t'} K(t_l, t)]_l$.

(f) For $K(t, t') = e^{-(t-t')^2/(2\tau^2)}$, compute $K_{vv}(t, t + \tau)$. Discuss.

[2 marks]

Solution

$$K_{vv}(t, t') = \frac{1}{\tau^2} (1 - (t - t')^2 / \tau^2) e^{-(t-t')^2/(2\tau^2)}.$$

At $t - t' = \tau$: $K_{vv} = 0$.

Velocities separated by τ are uncorrelated — velocity decorrelates on the same scale as position. For smoother velocity, use a higher-order kernel (e.g. Matérn with greater smoothness) or model velocity directly as a GP.

4 Topic 4: MLE / MAP Estimation

4.1 2017 Q2(b): MAP M-Step for HMM Transition Matrix [8 marks]

Consider an HMM with hidden states $s_t \in \{1 \dots K\}$, transition matrix $\Phi_{ij} = P(s_t=j \mid s_{t-1}=i)$, output distributions $A_i(x_t)$, forward/backward messages

$$\alpha_t(i) = P(x_1, \dots, x_t, s_t=i \mid \Phi, \{A_j\}, \pi),$$

$$\beta_t(i) = P(x_{t+1}, \dots, x_T \mid s_t=i, \Phi, \{A_j\}),$$

and prior $\Phi_{i\cdot} \sim \text{Dirichlet}(\lambda)$, $\lambda \in \mathbb{R}_+^K$.

Derive the MAP M-step update for Φ .

[8 marks]

Solution

Define expected transition counts $N_{ij} = \sum_{t=1}^{T-1} \xi_t(i \rightarrow j)$ where $\xi_t(i \rightarrow j) = \frac{\alpha_t(i)\Phi_{ij}A_j(x_{t+1})\beta_{t+1}(j)}{P(x_{1:T})}$. The MAP objective for row i is $\sum_j (N_{ij} + \lambda_j - 1) \log \Phi_{ij}$ subject to $\sum_j \Phi_{ij} = 1$. Using Lagrange multipliers:

$$\hat{\Phi}_{ij}^{\text{MAP}} = \frac{N_{ij} + \lambda_j - 1}{\sum_{j'} (N_{ij'} + \lambda_{j'} - 1)}.$$

The Dirichlet prior adds pseudocounts $\lambda_j - 1$ to each transition count.

4.2 2020 Q2 / 2022 Q2: MLE for Markov Chains [20 marks]

Markov chain with states $1, \dots, K$, transition matrix P (P_{ij} = prob. $X_{t+1} = i$ given $X_t = j$, columns sum to 1), transition counts N_{ij} .

(a) Show that $\min \sum_k f_k(x_k)$ s.t. $g_k(x_k) = 0$ separates into K problems when f_k are convex and g_k linear. [3 marks]

Solution

The Lagrangian $L = \sum_k [f_k(x_k) + \mu_k g_k(x_k)]$ decomposes into independent terms. KKT conditions for the joint problem reduce to KKT per sub-problem. Since each f_k is convex and g_k linear, KKT is necessary and sufficient, so solutions coincide.

(b) Write the log-likelihood and formulate MLE as a convex problem.

[6 marks]

Solution

Log-likelihood: $\ell(P) = \sum_{i,j} N_{ij} \log P_{ij}$. The convex problem is:

$$\min_P - \sum_{i,j} N_{ij} \log P_{ij} \quad \text{s.t.} \quad \sum_i P_{ij} = 1 \quad \forall j, \quad P_{ij} \geq 0.$$

This has separable structure $\sum_j f_j(P_{\cdot j})$ with column-wise constraints, matching (a).

(c) Solve for column j : show $\hat{P}_{kj} = N_{kj} / \sum_i N_{ij}$.

[8 marks]

Solution

Column j sub-problem: $\min - \sum_k N_{kj} \log P_{kj}$ s.t. $\sum_k P_{kj} = 1$. Lagrange multiplier:

$$-\frac{N_{kj}}{P_{kj}} - \lambda = 0 \implies P_{kj} = -N_{kj}/\lambda.$$

Enforcing $\sum_k P_{kj} = 1$: $\lambda = -\sum_k N_{kj}$, so $\hat{P}_{kj} = N_{kj} / \sum_i N_{ij}$.

The Hessian $\text{diag}(N_{kj}/P_{kj}^2)$ is positive definite, confirming the global minimum.

Now impose an invariant distribution π : $\mathcal{M}_\pi = \{\text{all chains with invariant distribution } \pi\}$.

(d) State MLE in $\mathcal{M}_\pi$ as a convex problem. Can (a) be used? [3 marks]

Solution

Add the constraint $\sum_j P_{ij}\pi_j = \pi_i$ for all i , plus $\sum_i P_{ij} = 1$ and $P_{ij} \geq 0$. The invariance constraint *couples* different columns of P , so part (a) **cannot** separate the problem. It must be solved jointly.

Shuhao Wang

5 Topic 5: MCMC / Sampling

5.1 2022 Q4: Markov Chains and Sampling Algorithms [20 marks]

Consider $\mathcal{K} = \{1, \dots, K\}$, functions $f_\theta : \mathcal{K} \rightarrow \mathcal{K}$, random parameters $\Theta_1, \dots, \Theta_n$, and

$$F_n(x) := f_{\Theta_1} \circ \dots \circ f_{\Theta_n}(x), \quad G_n(x) := f_{\Theta_n} \circ \dots \circ f_{\Theta_1}(x).$$

(a) With Θ_i generated i.i.d., from a Markov chain, or from an HMM — in which cases is $(G_1(x), \dots, G_n(x))$ a Markov chain? [3 marks]

Solution

i.i.d.: $G_{n+1}(x) = f_{\Theta_{n+1}}(G_n(x))$ with Θ_{n+1} independent of the past given $G_n(x)$. **Yes**, Markov.
Markov chain / HMM: Knowing $G_n(x)$ does not determine Θ_n , so Θ_{n+1} is not conditionally independent of the past given $G_n(x)$ alone. **Not** Markov in either case.

(b) (i.i.d. Θ_i) Do $F_n(x)$ and $G_n(x)$ have the same distribution? Do the sequences? [3 marks]

Solution

Marginals: Yes. $(\Theta_1, \dots, \Theta_n) \stackrel{d}{=} (\Theta_n, \dots, \Theta_1)$ for i.i.d., so $F_n(x) \stackrel{d}{=} G_n(x)$.
Sequences: No. (G_n) is Markov; (F_n) is not (all share f_{Θ_1} as outermost map, creating non-Markov dependencies). Joint distributions differ.

Suppose F_N becomes constant: $F_N(x) = X$ for all x , with $X \sim \pi$.

(c) What is the distribution of $F_{N+1}(x)$? [3 marks]

Solution

Composing $N + 1 \geq N$ i.i.d. maps still produces a constant function. By the i.i.d. property, the distribution is invariant: $F_{N+1}(x) \sim \pi$.

(d) Specify an algorithm (single while loop) that outputs a draw from π . [3 marks]

Solution

```
n <- 0
while F_n is not constant on K:
  n <- n + 1
  (draw Theta_n; extend F_n reusing Theta_1, ..., Theta_{n-1})
return F_n(x) // for any x in K
```

Constancy is checked by evaluating F_n on all $x \in \mathcal{K}$ (finite set).

(e) Which is preferable for a single draw from π : this algorithm or Metropolis-Hastings? [3 marks]

Solution

Propp–Wilson produces an **exact** sample from π with no burn-in bias. MH gives only approximate samples after finite burn-in, with correlated outputs. For a single exact draw, Propp–Wilson is preferable (assuming it terminates efficiently).

(f) (Staircase density sampler) We divide the area under a density p into three blocks. Starting from X_{i-1} : draw $T_i \sim \text{Uniform}[0, p(X_{i-1})]$, select the block k containing (X_{i-1}, T_i) , sample (X_i, Y_i) uniformly from block k , keep X_i . If $X_{i-1} \sim p$, is $X_i \sim p$? [5 marks]

Solution

Assume $X_{i-1} \sim p$. Then (X_{i-1}, T_i) is uniform under the graph of p . Block k is selected with probability $\text{Area}(k)/1$. Then (X_i, Y_i) is uniform on block k .

The marginal density of X_i :

$$p_{X_i}(x) = \sum_k \frac{\text{Area}(k)}{1} \cdot \frac{h_k(x)}{\text{Area}(k)} = \sum_k h_k(x) = p(x),$$

since the blocks partition the area under p , so $\sum_k h_k(x) = p(x)$ for all x .

Yes, this is a valid sampler: $X_{i-1} \sim p \implies X_i \sim p$.

6 Topic 6: Model Comparison / Marginal Likelihood

Covered within the Conjugate Mines question (Section 2.1, parts (f)–(h)). The technique is computing and comparing marginal likelihoods: the model with higher marginal likelihood is favoured (Bayesian model selection via Bayes factors).

Shuhan Wang

7 Topic 7: Mixture Models and EM

7.1 2019 Q2: A Mixture of Linear Experts [20 marks]

We are given pairs (x_n, y_n) , $n = 1 \dots N$, with gating variables $s_n \in \{1 \dots K\}$:

$$P(s_n=k | x_n, \Theta) = \pi_k(x_n) = \frac{e^{-\frac{1}{2}\|x_n - \mu_k\|^2}}{\sum_{k'} e^{-\frac{1}{2}\|x_n - \mu_{k'}\|^2}}, \quad P(y_n | s_n=k, x_n, \Theta) = \mathcal{N}(W_k x_n, \Psi),$$

where $\Theta = \{\mu_1 \dots \mu_K, W_1 \dots W_K, \Psi\}$.

(a) Show that $P(y_n, s_n | x_n, \Theta)$ is a joint exponential family model.

[6 marks]

Solution

Write s_n as a one-hot vector $\mathbf{s}_n$ with $[\mathbf{s}_n]_k = \delta_{s_n, k}$. Then:

$$\log P(y_n, \mathbf{s}_n | x_n) = \sum_k [\mathbf{s}_n]_k \left[\log \pi_k(x_n) - \frac{1}{2} \log |2\pi\Psi| - \frac{1}{2} y_n^\top \Psi^{-1} y_n + y_n^\top \Psi^{-1} W_k x_n - \frac{1}{2} x_n^\top W_k^\top \Psi^{-1} W_k x_n \right].$$

The sufficient statistics are $\{[\mathbf{s}_n]_k\}$ and $\{[\mathbf{s}_n]_k \cdot y_n\}$. The natural parameters are: for $[\mathbf{s}_n]_k$, $\eta_k = \log \pi_k(x_n) - \frac{1}{2} \log |2\pi\Psi| - \frac{1}{2} x_n^\top W_k^\top \Psi^{-1} W_k x_n$; for $[\mathbf{s}_n]_k y_n$, $\eta_{k,y} = \Psi^{-1} W_k x_n$. This is joint exponential family form with x_n -dependent natural parameters.

(b) Find $P(s_n=k | x_n, y_n, \Theta)$.

[4 marks]

Solution

By Bayes' rule:

$$r_{nk} = P(s_n=k | x_n, y_n, \Theta) = \frac{\pi_k(x_n) \mathcal{N}(y_n; W_k x_n, \Psi)}{\sum_{k'} \pi_{k'}(x_n) \mathcal{N}(y_n; W_{k'} x_n, \Psi)}.$$

(c) Derive the M-step update for W_k .

[4 marks]

Solution

From $\frac{\partial Q}{\partial W_k} = \sum_n r_{nk} \Psi^{-1} (y_n - W_k x_n) x_n^\top = 0$:

$$\hat{W}_k = \left(\sum_n r_{nk} y_n x_n^\top \right) \left(\sum_n r_{nk} x_n x_n^\top \right)^{-1}.$$

This is a responsibility-weighted least-squares solution.

(c, continued) Derive the M-step update for μ_k .

[6 marks]

Solution

The dependence on μ_k enters through $\log \pi_k(x_n)$. Computing the derivative:

$$\frac{\partial \log \pi_k(x_n)}{\partial \mu_k} = (1 - \pi_k(x_n))(x_n - \mu_k).$$

Setting $\frac{\partial Q}{\partial \mu_k} = \sum_n r_{nk} (1 - \pi_k(x_n))(x_n - \mu_k) = 0$:

$$\hat{\mu}_k = \frac{\sum_n r_{nk} (1 - \pi_k(x_n)) x_n}{\sum_n r_{nk} (1 - \pi_k(x_n))}.$$

This is a weighted average that upweights points where the gating function does not already strongly favour component k .

8 Topic 8: Nonlinear Regression / ARD (GP Connection)

8.1 2017 Q6: Nonlinear Regression [20 marks]

Consider:

$$\begin{aligned} y_i &= \beta^\top \phi(x_i) + \epsilon, & \epsilon &\sim \mathcal{N}(0, \sigma^2), \\ \beta &\sim \mathcal{N}(0, A^{-1}), & A &= \text{diag}[\alpha_1 \dots \alpha_K]. \end{aligned}$$

Collect $\phi(x_i)$ into Φ ($K \times N$) and y_i into row vector Y .

(a) Write $P(Y | \Phi, \beta, \sigma^2)$ as a vector normal on $Y^\top$.

[3 marks]

Solution

$$Y^\top = \Phi^\top \beta + \epsilon, \text{ so } P(Y^\top | \Phi, \beta, \sigma^2) = \mathcal{N}(Y^\top; \Phi^\top \beta, \sigma^2 I_N).$$

(b) Find the posterior on β .

[5 marks]

Solution

Completing the square in β from the log-posterior:

$$P(\beta | Y, \Phi) = \mathcal{N}(\mu_\beta, \Sigma_\beta), \quad \Sigma_\beta = (A + \sigma^{-2} \Phi \Phi^\top)^{-1}, \quad \mu_\beta = \sigma^{-2} \Sigma_\beta \Phi Y^\top.$$

(c) Find the marginal likelihood $P(Y^\top | \Phi, A, \sigma^2)$.

[3 marks]

Solution

$Y^\top$ is a linear transformation of Gaussian β plus Gaussian noise, so the marginal is Gaussian. By the law of total expectation and variance:

$$P(Y^\top | \Phi, A, \sigma^2) = \mathcal{N}(0, \Phi^\top A^{-1} \Phi + \sigma^2 I_N).$$

(d) Interpret ARD for this nonlinear model.

[3 marks]

Solution

Optimising over A drives some $\alpha_k \rightarrow \infty$, setting $\beta_k \rightarrow 0$ and removing feature ϕ_k . This is *automatic basis function selection*: irrelevant nonlinear features are pruned, rather than input dimensions as in linear ARD.

(e) Compute the EM M-step update for A .

[3 marks]

Solution

E-step: $P(\beta | Y) = \mathcal{N}(\mu_\beta, \Sigma_\beta)$. M-step: maximise $\langle \log P(\beta | A) \rangle_q$ w.r.t. α_k :

$$\hat{\alpha}_k = \frac{1}{\langle \beta_k^2 \rangle_q} = \frac{1}{[\Sigma_\beta]_{kk} + \mu_{\beta,k}^2}.$$

Alternative: fix the prior to $\beta \sim \mathcal{N}(0, I)$ and use $y_i = \beta^\top \phi(\gamma^{-1} \circ x_i) + \epsilon$ with D -dimensional hyperparameter γ . With the kernel $K(x_i, x_j) = e^{-\|x_i - x_j\|^2}$:

(f) Interpret γ within this model.

[3 marks]

Solution

The kernel becomes $K = \exp(-\sum_d (x_{id} - x_{jd})^2 / \gamma_d^2)$. Each γ_d is a **length-scale** for input dimension d . Optimisation drives irrelevant $\gamma_d \rightarrow \infty$, making the kernel insensitive to that dimension — implementing **automatic relevance determination at the input level**. The regression depends

only on dimensions with finite γ_d .

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